Data-Quality Assessment

AnalystLab Africa Experience Lab | Week 3 | Track A: Data Analytics | Author: Divine Okezie Nwaoha | Tools: Power BI, Power Query, DAX, Python/SQL (validation)

Data-Quality Assessment

Files used

File	Rows	Columns	Used for
encounters.csv	27,891	14	Volumes, classes, cost, coverage, duration
patients.csv	974	20	Patient demographics (identifiers removed)
procedures.csv	47,701	9	Procedure frequency and cost
payers.csv	10	7	Payer names
organizations.csv	1	6	Single hospital context

Relationships: Patients[Id] to Encounters[PATIENT]; Payers[Id] to Encounters[PAYER]; Encounters[Id] to Procedures[ENCOUNTER]; Organizations[Id] to Encounters[ORGANIZATION]; DimDate[Date] to Encounters[Encounter_Date].

Checks passed: all keys are unique where expected; no orphan patients, payers, organizations or procedures; no invalid timestamps; no STOP earlier than START; no full-row duplicates in any file.

Issues found

#	Issue	Size	Decision	Effect on analysis
1	Extreme encounter durations (longest about 44,930 hours)	75 encounters over 24h (0.27%)	Kept and flagged	Averages of duration would be distorted; the Over 24h % KPI is used instead
2	Zero payer coverage ambiguous	13,586 encounters; 8,807 are NO_INSURANCE, about 4,779 insured	Kept; analysed by payer	Cannot assume "unfunded"; five private insurers record about 0% coverage
3	Zero total claim cost	89 encounters	Kept	Slightly lowers average claim cost
4	Partial years	2022 has 220 encounters (data ends 5 Feb); 2011 ramp-up	Flagged on dashboard	Yearly trend must not be read as decline
5	Possible duplicate encounters	27 share patient, start time and class (different IDs)	Kept	Could be legitimate multiple events; effect is under 0.1%
6	Procedure codes vs descriptions	157 codes, 163 descriptions	Grouped by code	Avoids splitting one procedure
7	Procedure base cost skew	Range from 1 to 289,531	Median shown; minimum-volume slider	Raw averages mislead for rare procedures
8	Column names UPPERCASE vs dictionary Title Case	All files	Documented	None

#	Issue	Size	Decision	Effect on analysis
9	Overlapping encounters	560 encounters start before the previous one ends	Documented	30-day return calculation treats these as same-episode
10	Personal identifiers present	Names, addresses, coordinates	Removed in Power Query; not shown in any visual	Privacy

Metrics that could mislead

- **Zero Coverage %:** may reflect missing capture, not missing funding.
- **Average claim by payer:** includes NO_INSURANCE as a "payer" and 89 zero-cost claims.
- **Average procedure cost:** driven by rare, very expensive procedures unless a volume floor is set.
- **Yearly encounter trend:** 2011 and 2022 are partial or ramp-up periods.
- **30-day return rate:** any later encounter, not a clinical readmission (no diagnosis or discharge data).